

Yingrui Hou

Ingénieure modélisation, simulation & validation pour systèmes complexes

Target Application: Capgemini Engineering | Ingénieure/Ingénieur
Modélisation Et Simulation

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Introduction personnelle

Ingénieure modélisation et validation avec une base forte en physique expérimentale et en workflows C++/Python pour systèmes physiques complexes.

J'interviens là où simulation, données d'essai et validation doivent être reliées proprement pour produire des modèles crédibles et exploitables.

Mon apport principal est la structuration de workflows de simulation, calibration, analyse et validation dans des environnements collaboratifs exigeants.

Selected Fit For This Role

- Developed reusable C++/Python workflows for modelling, validation, and inference on noisy systems without direct ground truth.
- Developed and adapted Geant4/C++ simulation code for system-response studies, with analysis outputs usable for technical assessment.
- Combined test measurements, simulation, and response maps to quantify the impact of physical non-idealities on system performance.
- Developed calibration and correction workflows to compare performance before and after adjustment and make models more reliable.
- Built validation and monitoring workflows that turned observed discrepancies into software or modelling corrective actions.
- Worked in large shared technical environments with review discipline, code reusability, and support for calibration and modelling decisions.

complex-system modelling, test integration, V&V, user support, and work in broad industrial engineering environments.

Education

PhD in Particle Physics
University of Chinese Academy of Sciences
BSc in Applied Physics
China University of Mining and Technology

Languages

Chinese: Native speaker
English: C1
French: B1 to B2

Core Tools & Methods

Tools: Geant4 | Python | Geant4 simulation | NumPy | statsmodels | C/C++ | ROOT / RooFit | Workflow Automation (YAML / Shell / CMake) | Framework Integration | pandas | etc

Methods: Statistical Inference | correlation-based correction | Probabilistic Modelling | Large-Scale Data Analysis | Detector Simulation | event-level correction | production yield workflows | Maximum Likelihood Estimation | Latent-Parameter Estimation | beam-test analysis | etc

Validation: Calibration & Bias Correction | Model Validation | Simulation-to-Data Alignment | Simulation Calibration | Simulation Correction | Monitoring | Data Validation | Bias Correction | Calibration | Statistical Validation | etc

ML: Domain Adaptation | Feature Validation | train-test diagnostics | scikit-learn | CatBoost / Gradient Boosting | GBDT | CatBoost | Reweighting | Feature Selection | CatBoost training | etc

Relevant Experience

Postdoctoral Researcher | 2021 - 2025

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.

Selected Project

Sensor Simulation & Reconstruction

Signal Processing | Simulation Calibration | Digital Twins

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor...

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction

Impact: linked measured non-uniformity to system-level resolution through both simulation code and experimental validation.

Industry relevance: Sensor Analytics | Measurement Systems | Simulation-Assisted Design

Test-Data Validation & Root-Cause Analysis

Monitoring | Data Validation | Performance Diagnostics | Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Impact: translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias.

Industry relevance: Production Monitoring | Model Validation | Observability | Quality Assurance

Pulse-Shape Discrimination for Energy Reconstruction

Pulse Shape Discrimination | Detector Simulation | Time-Resolved Signal Analysis | Resolution Optimisation

Built a PSD-oriented simulation and analysis workflow to extract shower composition from scintillation time profiles and use it to improve reconstructed energy.

Methods: particle-dependent timing models, component fitting, correlation-based correction

Impact: showed that PSD-derived shower fractions can improve reconstructed energy and help judge which detector materials remain usable for timing-based...

Industry relevance: Time-Resolved Sensing | Signal Decomposition | Measurement-System Design